

1. ABSTRACT

Sarees are traditional Indian garments available in different colours, patterns, and designs. Identifying saree designs manually can be difficult when many different patterns and colours are present.

This project presents a simple **Color-Invariant Saree Design Recognition** system using deep learning. The main objective is to recognize the design of a saree while reducing the influence of its colour.

The system uses saree images belonging to five design categories: **Floral, Geometric, Paisley, Plain, and Striped**. A pretrained **MobileNetV2** model is used through transfer learning for image classification. Image augmentation techniques are used during training to expose the model to different image conditions and colour variations.

The user can upload a saree image through a simple Streamlit web application. The system processes the image and predicts the saree design along with a confidence score.

The main purpose of this project is to demonstrate that saree design recognition can focus on visual patterns rather than depending only on colour.

2. INTRODUCTION

Artificial Intelligence and Deep Learning are widely used for image recognition and classification. Image classification allows a computer to identify the category of an object from an image.

Sarees contain many different visual characteristics such as colour, texture, borders, motifs, and patterns. The same saree design can appear in different colours. Therefore, a recognition system should ideally identify the design without depending strongly on colour.

For example:

Red Floral Saree → Floral

Blue Floral Saree → Floral

Green Floral Saree → Floral

The colour changes, but the design remains floral.

This project focuses on developing a simple deep-learning system for this purpose.